ChatGPT

Volatility Contraction → Expansion in Small-Cap Momentum Trading

Executive summary

Volatility Contraction → Expansion (VCE) is the observable transition from **shrinking price ranges and often declining volume** into a **range expansion event** (breakout, breakdown, or fast mean-reversion). In trading literature, the same idea appears under multiple names: **Bollinger "Squeeze"** (bandwidth narrowing), **John Carter's TTM Squeeze** (Bollinger Bands inside Keltner Channels), **Mark Minervini's Volatility Contraction Pattern (VCP)** (sequential tightening + volume dry-up into a pivot breakout), and **Wyckoff accumulation/markup** (absorption of supply before a run). The terminology differs, but the mechanism is consistent: when **order-flow aggressiveness temporarily falls**, the market becomes "coiled," and the next meaningful imbalance tends to produce a visibly larger move. [1]

In **small caps**, VCE behavior is often **amplified** by thin order books, wider spreads, and frequent **volatility interruptions** (Limit Up/Limit Down halts). A "quiet" tape in a low-float name can quickly become a violent expansion when new liquidity arrives (news, momentum scanners, social attention) or when supply is forced (short covering). [2]

Evidence quality is **asymmetric**:
- The existence of **volatility clustering** and persistence is one of the most robust statistical "stylized facts" in finance. [3]
- The claim "a volatility contraction predicts a profitable upside breakout" is **much less settled**. Academic work supports that some technical rules and chart patterns can have statistical predictability (e.g., long-horizon index tests; algorithmic pattern recognition), but profitability is heavily sensitive to **transaction costs, data-snooping, and regime changes**. [4]
- For small firms, predictability may be higher, but net profitability can still disappear once realistic costs are included—an important warning for small-cap traders where costs/ slippage are often large. [5]

Professionals use VCE in **two main ways**: 1) **Momentum continuation**: wait for contraction (tight base / micro-pullback) then enter on confirmed expansion with volume and quick feedback (e.g., "micro pullback" and "gap-and-go" style tactics). [6]
2) **Structure-based dip buys / mean-reversion into accumulation**: treat the contraction as **absorption** and take entries near defined risk points (trendline/VWAP/base low) or on "spring/test" behavior. [7]

Small caps add a decisive third dimension: **financing/dilution risk**. "ATM" programs and shelf capacity can turn apparent accumulation into a trap because new shares can be "trickled" into strength, capping expansion. Detecting this requires reading filings (Form S-3 eligibility, prospectus supplements, ATM distribution agreements) and monitoring EDGAR. [8]

Research scope and methodology

This report synthesizes **technical analysis literature**, **professional trader documentation**, **regulatory primary sources**, and **academic finance/microstructure research**.

Source types prioritized - Primary / official: U.S. Securities and Exchange Commission [9] guidance on Form S-3 eligibility and EDGAR search, and primary law/regulation defining “at-the-market offering” in Rule 415. [10]
- Peer-reviewed / archival academic papers and institutional working papers (e.g., ARCH/GARCH; stylized facts; technical rules; data-snooping controls; liquidity/float studies). [11]
- Reputable practitioner education with explicit rule sets and charts (e.g., Warrior Trading [12] educational articles describing entries/stops, and chart-pattern references like Nasdaq [13] pattern write-ups). [14]
- Secondary trader blogs and commercial tool descriptions were used only when (a) they clearly summarize a known framework and (b) they do not claim “proven win rates” without empirical backing; such claims are flagged as weaker evidence. [15]

Time range - Historical lineage sources span late 1800s (technical analysis origins attributed to Dow) through early 1900s (Wyckoff), and modern empirical finance (1980s onward). [16]
- Empirical/academic samples referenced include multi-decade datasets (DJIA 1897–1986; U.S. stocks 1962–1996; cross-country firm-years 2003–2011). [17]

Search strategy Search terms were grouped into five baskets:
- “volatility contraction pattern”, “VCP Minervini”, “line of least resistance”, “volatility dry up” [18]
- “Bollinger squeeze”, “Bandwidth indicator”, “TTM Squeeze Bollinger Keltner” [19]
- “small cap micro pullback”, “gap and go opening range breakout”, “day 2 continuation” [20]
- “volatility clustering”, “ARCH Engle”, “GARCH Bollerslev”, “stylized facts Cont” [3]
- “ATM offering Rule 415”, “Form S-3 one-third public float”, “EDGAR search at-the-market” [21]

Origins and conceptual lineage

The core “contraction then expansion” idea predates modern indicators. What changed over time is **measurement** (bands/ATR), **language** (squeeze/VCP), and **market structure** (speed, fragmentation, halts).

Early charting and campaign theory - Technical analysis is described in academic literature as “dating back to the 1800’s,” with some of the oldest techniques attributed to Charles Dow. [22]
- Richard D. Wyckoff [23] framed markets as campaigns with phases: **accumulation → markup → distribution**, emphasizing that meaningful “swings” often begin while accumulation/distribution is visible in the preparatory stage. [24]
This is an early conceptual ancestor of VCE: *absorption (tightening) before markup (expansion).* [24]

Statistical finance formalizes volatility clustering - Volatility clustering—high volatility tending to cluster and persistence in volatility measures—is a widely-documented stylized fact. [3]
- ARCH/GARCH models formalize time-varying conditional variance, supporting the idea that volatility regimes change and persist. [25]
Important nuance: these models support **predictability of volatility**, not necessarily **directional predictability of returns**. [26]

Indicator era: squeezes and band contraction - John Bollinger [27] 's Bollinger Bands popularized visual volatility expansion/contraction; "squeeze" setups focus on periods where bands tighten and then expand again. [28]
- StockCharts.com [29] formalized related education around Bollinger/TTM squeeze heuristics (Bollinger Bands compressed inside Keltner Channels) and the "squeeze firing." [30]

Growth-stock literature: tight bases and rare momentum structures - William J. O'Neil [31] 's "High Tight Flag" is described as a rare, powerful pattern: a rapid doubling followed by a relatively shallow pullback (often 20–25%). [32]
This is another VCE variant: extreme expansion (flagpole), contraction (flag), then expansion (breakout). [33]

Modern naming: Minervini's VCP - The best-known branded "volatility contraction" framework is Mark Minervini [34] 's VCP, described as his "signature stock setup" in an interview. [35]
- Many contemporary educator sources attribute the VCP label to Minervini and describe its elements as sequentially smaller pullbacks plus volume dry-up into a pivot breakout. [36]
Evidence note: the *label* and specific "VCP" rule-language are modern; the underlying *compression → breakout* concept is much older (Wyckoff/Dow/Bollinger). [37]

Professional playbooks and documented strategies

VCE shows up in professional playbooks in **two families**: continuation breakout entries and structural dip buys into absorbed supply. The difference is *where you enter relative to the contraction* and *how you define invalidation*.

Minervini VCP

Core idea: a stock in an uptrend forms a base where **each pullback is smaller** and volume contracts, creating a "line of least resistance" into a pivot. [18]

Rule-like elements commonly documented: - **Context filter**: VCPs are treated as continuation patterns and are typically sought in existing uptrends. [38]
- **Contraction structure**: 2–6 increasingly smaller pullbacks; the tightest area is near the pivot. [38]
- **Volume behavior**: volume "dries up" during contractions; breakout occurs on expansion (relative to recent average). [38]
- **Entry trigger**: break above pivot/upper boundary with decisive expansion in volume and range. [38]
- **Stop logic**: stop below the last contraction low or a volatility-based distance (e.g., ATR multiple) depending on instrument and timeframe. Practitioner education emphasizing ATR-based stops is common. [39]

Small-cap adaptation: VCP logic can be applied to **intraday bases** after a catalyst spike (tight 5-min base into continuation) and to **Day-2/Day-3 consolidations** where daily range compresses before the next leg. This "intraday VCP" approach has been described publicly by competition-level traders as tightening pullbacks around a short EMA with declining volume before a breakout. [40]

Evidence quality: Minervini is a widely cited practitioner, but public "win rate" claims for VCP breakouts are usually **educator assertions** rather than peer-reviewed statistics; treat them as hypotheses to test. [41]

Bollinger Squeeze and TTM Squeeze

Bollinger "squeeze" heuristics: - A squeeze visually appears when Bollinger Bands **tighten markedly** during consolidation; after compression, price often makes a larger move "in either direction," ideally with volume confirmation. [42]
- The setup is **direction-agnostic**; you need a directional trigger (band break, breakout from structure, momentum confirmation). [43]

TTM Squeeze logic (Bollinger inside Keltner): - A squeeze is identified when Bollinger Bands are **completely enclosed** within Keltner Channels; when bands expand back outside, the squeeze "fires." [44]
- This is explicitly framed as **volatility compression** with a likely upcoming expansion, but not a guaranteed directional edge. [45]

Ross Cameron's momentum "micro-contraction → expansion" entries

Small-cap momentum traders often trade **micro VCE** on the 1-minute/5-minute chart: an active mover pauses briefly (micro contraction), then expands again.

Ross Cameron [46] documents multiple setup rulesets consistent with VCE logic: - Micro pullback scalping: require the stock to "already be moving," wait for a "micro pullback" (single red candle/tight consolidation), then enter as the pivot reclaims, using tight stops. [47]
- Momentum pullback logic: "first green candle to make a new high after the pullback" as entry; stop at the low of pullback; explicit tight risk anchoring (e.g., 20 cents) is discussed as a way to preserve favorable payoff ratios. [48]
- Opening range / "Gap and Go" definition: buy the 1-minute opening range breakout with a stop at the low, emphasizing catalyst + interest + float. [49]

These are "VCE-like" because the first pause after expansion is treated as information: if the pause is **tight** (no heavy supply shows), the next expansion is more tradable; if the pause deepens or becomes choppy, the thesis is invalidated quickly. [50]

Wyckoff spring/test as a "liquidity trap → expansion"

Wyckoff's accumulation framework describes how campaigns often begin with a preparatory stage (accumulation), followed by markup. [51]
In modern Wyckoff education, a "spring" is often described as a downside probe (false breakdown) that tests demand before an advance, frequently followed by a "test" on lighter volume. [52]

In small caps, you can see analogous behavior around a **prior day low / VWAP / key bid**: price briefly breaks, triggers stops, then reclaims, and volatility expands upward as trapped sellers cover. This is conceptually consistent with Wyckoff's supply test principles. [53]

Roland Wolf's small-cap approach

Documentation available in text form is thinner than for Minervini/Cameron, but interviews and summaries emphasize: - A focus on **overnight holds** under PDT constraints (multi-day continuation logic). [54]
- Increased attention to **VWAP** and adapting patterns post-2020 as market participation changed. [55]

This aligns with a VCE read: he appears to prefer waiting for the **secondary spike** after structure forms (i.e., confirmation expansion after contraction) rather than chasing first-move breakouts. [56]

Setup comparison table

Setup label	Typical timeframe	"Contraction" definition	Primary trigger	Stop anchor	Best fit in small caps	Key failure mode
VCP (Minervini-style)	Daily → weekly; adaptable to 5-min	Sequential smaller pullbacks + volume dry-up	Break above pivot with volume expansion [38]	Last contraction low / base low	Day-2/ Day-3 tight base; intraday "tight coil" after catalyst [40]	False breakout + overhead supply/ dilution
Bollinger "Squeeze"	Any	Bandwidth tightens; volatility minimal [42]	Break of band + structure	Outside consolidation	Broad applicability	Directionless signals; whipsaws
TTM Squeeze	Any	BB inside KC; "fires" when BB exit [30]	"Fire" + momentum	Outside squeeze range	Clean visual scanner filter	Fires into chop if no catalyst
Micro pullback (momentum)	1-min/ 5-min	One-candle/ tight pause in a mover [47]	Pivot reclaim/ new high	Micro low	Premarket gappers; low float	Liquidity vacuum; halts/ reversals
Wyckoff spring/test	5-min to daily	False breakdown + reclaim; test on lighter volume [52]	Reclaim + SOS	Under spring low	"Liquidity trap → expansion" days	Real breakdown (no reclaim)
High Tight Flag	Daily	Large run + shallow consolidation [32]	Breakout above flag	Under flag low	Rare but explosive in hot tape	Exhaustion / blow-off top

Evidence from market microstructure and academic research

This section separates what is **well supported** (volatility regimes, liquidity effects) from what is **contested** (profitable breakout edges).

Volatility clustering supports “expansion after quiet” but not direction

A canonical stylized fact: volatility measures show **positive autocorrelation**—high-volatility events cluster in time. [26]
ARCH and GARCH models formalize this as **time-varying conditional variance**. [25]

Interpretation for VCE: contraction periods can reasonably be treated as **low-volatility regimes**, and regime changes can produce expansions. However, volatility clustering alone does **not** imply that expansions are predictably bullish; it mainly supports that volatility itself is forecastable to some extent. [26]

Technical rule predictability is documented; robustness is debated

Long history index tests: - The Santa Fe Institute [57] working paper by Brock–Lakonishok–LeBaron tests moving average and trading range break rules on the DJIA (1897–1986) and reports that buy periods have higher returns than sell periods; an illustrative example given is buy-period daily return of 0.042% vs sell-period −0.025%. [22]
- Sullivan–Timmermann–White explicitly focus on **data-snooping**: evaluating a universe of rules and correcting inference with White’s Reality Check. They apply rules to 100 years of DJIA data and quantify the dangers of “finding” profitable rules by trying many. [58]
- White defines data snooping as reusing the same dataset for model selection/inference, risking results that are “simply due to chance.” [59]

Pattern recognition (systematizing “chart reading”): - Andrew W. Lo [60] and coauthors propose algorithmic technical pattern recognition and apply it to a large number of U.S. stocks from 1962–1996 to evaluate effectiveness, explicitly targeting the subjectivity critique (“eyes of the beholder”). [61]

Meta-level surveys: - Park & Irwin’s comprehensive review summarizes both practitioner usage and empirical studies, noting that 30–40% of practitioners (in surveyed contexts) view technical analysis as important up to ~6 months, and cataloging 92 “modern” studies with mixed outcomes (more positive than negative results) while emphasizing methodological issues like data-snooping and transaction cost estimation. [62]
- Their later journal survey abstract similarly notes early vs modern study differences and differing profitability by market type (FX/futures vs stocks). [63]

Small firms: higher predictability, harder profitability

A paper focused on company size finds that technical rules have “progressively higher predictive ability the smaller the size of the company,” yet still “are not profitable given transaction costs” (UK sample of 100 stocks, 1987–2002, per the abstract). [64]
This matters for small-cap VCE because **slippage, spread, borrow costs, and halts** can dominate theoretical edge. [65]

Liquidity and float: why small-cap VCE can be violent

Liquidity/price impact: - Yakov Amihud [66] shows an illiquidity measure (absolute return per dollar volume) interpretable as price response to volume and reports strong effects across NYSE stocks (1964–1997), with effects stronger for small firms’ stocks. [67]

Free float and liquidity: - Ding–Ni–Zhong study 110,562 firm-year observations across 55 countries (2003–2011) and finds stock liquidity increases with free float; higher free float is associated with lower liquidity risk. [68]
Small-cap inference: very low float often means lower natural liquidity and larger price impact, so the shift from contraction to expansion can be abrupt. [69]

Volatility halts: - Limit Up/Limit Down mechanisms can pause individual stocks when price swings exceed bands. [70]
For VCE traders, halts create a second "expansion channel" (gap risk) that is not present in many large-cap backtests. [71]

Dilution and issuance: a structural reason expansions fail

Academic issuance results: - Tim Loughran [72] and Jay Ritter [73] document that firms issuing stock (1970–1990) underperform in the long run; the paper's abstract reports average returns of ~5%/year for IPO firms and ~7%/year for SEO firms over five years after issuance, and details a large sample including 3,702 SEOs. [74]

Microstructure-relevant issuance mechanics: - An ATM is a continuous offering selling newly issued shares into the market via an agent "at prevailing market prices," often designed to "trickle" shares with minimal market impact. [75]
- U.S. regulation defines "at the market offering" in Rule 415. [76]
- SEC guidance describes Form S-3 eligibility for smaller companies and the one-third public float limitation ("baby shelf"). [77]

Small-cap trading implication: VCE setups can fail not because structure was "wrong," but because **new supply** can enter during the expansion phase, absorbing demand and reversing the move.

Algorithmic implementation and backtesting design

This section translates VCE from "pattern talk" into measurable rules. The goal is not to claim a universal edge but to define a **testable** framework.

Measurable definitions

VCE can be expressed using three measurable components:

Contraction - Range contraction: rolling true range or high-low range decreases over N bars.
- ATR contraction: ATR(14) today is low relative to its own history (e.g., ATR percentile).
- Bandwidth contraction: Bollinger Bandwidth or Keltner Channel width in bottom percentile of lookback. [78]

Participation decay - Volume dry-up: volume below moving average and/or decreasing through the contraction window (a common criterion in VCP education). [38]
- RVOL filter: relative volume above a threshold *at breakout* (not necessarily during contraction) is often used by momentum traders. [50]

Expansion trigger - A range expansion bar (wide candle), often with volume expansion. [79]
- A structural break (pivot high / prior day high / consolidation boundary). [80]

Scanning rule templates

Below are two rule “templates” (daily swing and intraday momentum). Thresholds should be treated as starting points and optimized cautiously due to data-snooping risk. [81]

Daily VCE scanner template

Universe filters (small-cap friendly) - Price > $1 (reduce micro-penny distortions)
- Avg dollar volume (20d) above your minimum liquidity
- Optional: float / market cap constraints (if you have reliable float data), knowing low float increases volatility.

Contraction - ATR(14) percentile over 252 days < 20th percentile
- OR Bollinger Bandwidth percentile < 20th percentile [82]
- AND last 10-bar high-low range < 1.5× ATR(14) (tight base proxy)

Structure - Close within X% of a resistance/pivot (e.g., within 5% of 20-day high)
- Uptrend filter if using continuation logic (moving averages aligned) [83]

Breakout watch - Alert when price breaks above pivot on volume > 1.5× 20-day avg volume (common breakout heuristic) [83]

Intraday VCE scanner template for small-cap momentum

Universe - Stocks already “in play”: large percent move, catalyst, high RVOL. [20]

Contraction - After initial push, detect a “micro base”:
- last 5 bars have lower average true range than prior 5 bars
- and highs are compressing into a level (flat top / wedge)

Trigger - Entry on reclaim/pivot break (new high of micro base) with tape/volume surge; stop at micro low. [84]

Pseudocode for a VCE scanner

```
def vce_daily_scan(bars, lookback=252):
    """
    bars: DataFrame with columns: open, high, low, close, volume indexed by date
    returns: True if VCE conditions met (contraction + structure), else False
    """
    # --- indicators ---
    atr14 = ATR(bars['high'], bars['low'], bars['close'], n=14)
    bb_width = bollinger_bandwidth(bars['close'], n=20, k=2)  # (upper-lower)/
```

```python
mid
    vol20 = bars['volume'].rolling(20).mean()

    # --- contraction ---
    atr_pct = percentile_rank(atr14.iloc[-1], atr14.tail(lookback))
    bbw_pct = percentile_rank(bb_width.iloc[-1], bb_width.tail(lookback))
    contraction = (atr_pct < 0.20) or (bbw_pct < 0.20)

    # --- tight base proxy ---
    base_range_10 = bars['high'].tail(10).max() - bars['low'].tail(10).min()
    tight_base = base_range_10 < 1.5 * atr14.iloc[-1]

    # --- volume dry-up proxy ---
    vol_dry = bars['volume'].tail(5).mean() < 0.8 * vol20.iloc[-1]

    # --- structure: near pivot ---
    pivot = bars['high'].rolling(20).max().iloc[-2]  # prior 20d high
    near_pivot = bars['close'].iloc[-1] >= 0.95 * pivot

    return contraction and tight_base and vol_dry and near_pivot


def vce_breakout_trigger(today_bar, pivot, avg_vol20):
    """
    breakout confirmation: close > pivot AND volume expansion.
    """
    return (today_bar.close > pivot) and (today_bar.volume > 1.5 * avg_vol20)
```

Backtest design suggestions for small caps

A credible VCE backtest for small caps must explicitly address:

- **Survivorship bias**: use historical constituents and delisted tickers.
- **Corporate actions**: reverse splits, symbol changes, halts.
- **Fill assumptions**: use realistic spreads and slippage; consider that range expansion bars often have worse fills (a point noted even in breakout education). 39
- **Event risk**: LULD pauses and gaps should be modeled (e.g., worst-case gap through stop). 70
- **Dilution regimes**: incorporate an issuance risk proxy (active shelf/ATM) and compare performance “with vs without” that filter. 85
- **Multiple testing control**: walk-forward evaluation and/or reality-check style controls when optimizing thresholds. 81

Performance expectations (evidence-aware): - You should expect VCE strategies to show **strong regime dependence**: strong trending markets and “hot tape” periods tend to help continuation; choppy markets increase whipsaws. Park & Irwin emphasize that conclusions vary across markets and time and that methodology matters. 62
- Small-cap implementations are likely to have **higher gross volatility and higher cost drag** (spread/

slippage), aligning with research that trading profitability can vanish once costs are included even when predictive ability exists. [5]

Variations, failure modes, and practical trade framework

When to use which "VCE variation"

A practical mental model is: **what created the contraction?** - **Absorption contraction** (accumulation): price holds a level while volume/volatility dries up → continuation more likely if catalyst remains active. [86]
- **Exhaustion contraction** (after a blow-off): volatility shrinks because participation disappears → breakdown more likely. [39]
- **Synthetic contraction** (illiquidity): volatility shrinks because there are no orders → expansion can be random and fill quality poor. [87]

The most common small-cap VCE failure modes

Dilution / distribution disguised as "tightness" - An ATM program can sell shares "into the natural trading flow," creating the illusion of absorption while supply is continuously added. [88]
- Smaller issuers may use Form S-3 subject to baby shelf constraints; knowing whether a company is eligible and how much capacity remains matters for risk. [89]

False compression - A narrowing range without volume "truth" can simply be inactivity; breakouts then fail because there is no stored demand. [90]

The "breakout trap" - Many breakouts fail; one education source cites a Bulkowski failure-rate study and states failure rates have risen as markets modernized, emphasizing the need to distinguish high-probability cases. [39]

Halt risk and gap risk - LULD pauses are a mechanical expansion of risk: you may not be able to exit at your stop, and resumption can gap. [70]

Detecting dilution risk before it breaks the trade

A practical EDGAR workflow: 1) Use SEC EDGAR search (company filings / full-text) to search the ticker/CIK and scan for "at-the-market," "equity distribution," "prospectus supplement," "S-3," "424B," "F-3," "Wainwright," etc. [91]
2) Check S-3 eligibility and baby shelf constraints (one-third of float). [77]
3) If an ATM exists, read the distribution agreement language—ATMs explicitly reference Rule 415 and describe continuous selling into the trading market. (Example SEC filing language shows shares may be sold into trading markets and describes agent compensation.) [92]

A simple decision tree for entering or skipping

```
flowchart TD
A[Stock on watchlist\n(catalyst or relative strength)] --> B{Liquidity workable?
```

```
\nDollar volume, spreads, halts risk}
B -- No --> X[Skip / paper trade only]
B -- Yes --> C{Issuance / dilution risk high?\nActive shelf/ATM, frequent
offerings}
C -- Yes --> D{Is setup very high quality\nwith tight, fast invalidation?}
D -- No --> X
D -- Yes --> E[Trade smaller size\nor require stronger confirmation]
C -- No --> F{Is volatility contracting\ninto a defined level?}
F -- No --> X
F -- Yes --> G{Accumulation signs?\nHigher lows, volume dry-up,\nreclaims VWAP/
levels}
G -- No --> H[Only trade breakouts\nwith strong volume + fast exit]
G -- Yes --> I{Entry style}
I --> I1[Breakout entry:\nabove pivot with volume]
I --> I2[Dip/structure entry:\nVWAP/base low risk-defined]
I1 --> J[Stop: under last contraction\nor volatility-based]
I2 --> J
J --> K[Manage:\nScale/targets + time stop]
K --> L{Failure signs?}
L -- Yes --> M[Exit quickly]
L -- No --> N[Trail / take partials]
```

A lifecycle timeline for small-cap VCE setups

```
timeline
  title Small-cap VCE lifecycle (typical)
  Catalyst/Attention : PR/news/sector sympathy -> RVOL spikes -> spreads widen
  Initial Expansion : impulse leg -> halts possible -> breakout chasers enter
  Shakeout / Liquidity Trap : quick flush / stop run -> reclaim key level
  Contraction : ranges tighten -> volume dries up -> "in-play" base forms
  Expansion : breakout or continuation -> momentum feedback loop
  Distribution / Dilution Risk : heavy supply / ATM selling -> failed breakout -
> fade
```

Practical checklists and sample trade plans

VCE watchlist filters (small caps) - "In-play" requirement: catalyst + participation (RVOL). [20]
- Structure: contraction into an obvious level (prior day high, premarket high, pivot, VWAP). [93]
- Risk-gate: LULD sensitivity (expect pauses), and dilution risk scan. [94]

Sample trade plan: Day-2 continuation from contraction - Context: day-1 spike + day-1 close strong; day-2 opens above/near day-1 VWAP and begins a tight range (inside-day / tight base).
- Entry: break of day-2 base high (or reclaim of day-1 high) with volume expansion.
- Stop: under base low or under VWAP reclaim point (whichever is closer and defines invalidation).
- Targets: prior high(s), whole/half dollar liquidity levels (common small-cap behavior), plus time-based exit if expansion doesn't follow quickly. [50]

Sample trade plan: Intraday micro-VCE ("micro pullback") - Context: stock already moving, strong RVOL, clear reason, clean momentum. [20]
- Contraction: one red candle or tiny consolidation.
- Entry: pivot reclaim / micro high break.
- Stop: micro low (tight).
- Exit: into the next psychological level; bail on hesitation/absorption. [47]

Sample trade plan: Wyckoff-style spring (liquidity trap) - Context: defined support from prior base; breakdown quickly reclaims.
- Entry: reclaim + "test" holds (lighter selling).
- Stop: under spring low.
- Target: mid-range then range high; scale if thin liquidity. [52]

Seven "early failure" warnings for VCE breakouts (small-cap adapted) - Expansion bar breaks out on **low volume** (no participation). [39]
- Immediate reclaim failure: price "flutters" back into the base after breakout (lack of real shift). [39]
- Aggressive supply at the pivot (repeated hard rejections). [95]
- Volume increases on down candles inside the base (distribution signature). [96]
- Increasing halt frequency + weaker post-halt bids (momentum decaying). [70]
- Filing/financing overhang: active ATM/shelf and history of selling into strength. [85]
- Wider spreads/illiquidity at the exact breakout moment (signal may be real, but fills make R/R invalid). [87]

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